

HIGHVALE, AB, Canada

WMO: 712410

Lat: 53.449N

Lon: 114.464W

Elev: 2450

StdP: 13.44

Time Zone: -7.00 (NAM)

Period: 94-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.3	-14.6	-25.6	1.4	-20.2	-19.1	2.1	-14.3	26.4	32.5	22.4	19.8	5.2	260	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	81.7	65.3	78.4	63.5	75.4	61.7	67.9	77.7	66.0	75.4	64.0	72.4	7.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	99.9	72.8	62.5	92.7	70.2	60.7	86.7	68.2	33.8	77.4	32.1	75.5	30.6	72.5	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.9	21.0	17.9	DB	-28.4	87.6	6.7	3.6	-33.1	90.2	-37.0	92.3	-40.8	94.3	-45.6	96.9	(4)
(5)			WB	-27.2	75.0	7.7	6.6	-32.8	79.7	-37.3	83.5	-41.6	87.2	-47.3	92.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.7	12.2	20.5	26.5	41.5	51.0	58.4	63.4	61.4	53.0	41.5	27.1	18.4		
	DBStd	20.53	17.04	13.76	14.37	9.71	8.08	5.49	5.33	6.04	7.15	8.56	11.63	15.74		
	HDD50	5099	1171	826	729	278	85	4	1	4	50	283	687	981		
	HDD65	9334	1636	1246	1194	704	436	207	94	143	362	730	1137	1445		
	CDD50	1327	0	0	0	23	116	256	415	358	140	18	1	0		
	CDD65	86	0	0	0	0	2	9	43	31	1	0	0	0		
	CDH74	954	0	0	0	1	46	133	419	306	46	3	0	0		
	CDH80	178	0	0	0	0	5	23	88	58	4	0	0	0		
	WSAvg	7.7	6.4	7.5	8.2	8.6	8.4	7.9	7.5	6.6	7.6	8.0	7.5	8.0		
	PrecAvg	19.3	1.0	0.6	0.7	1.2	2.2	3.4	3.7	2.5	1.5	0.9	0.8	0.7		
(15) Precipitation	PrecMax	25.7	3.0	1.2	1.4	2.2	4.3	6.9	8.2	5.4	3.4	3.0	2.4	1.9		
	PrecMin	12.6	0.1	0.0	0.1	0.2	0.7	1.0	1.8	1.2	0.4	0.2	0.2	0.2		
	PrecStd	3.1	0.7	0.3	0.3	0.6	0.9	1.3	1.5	0.8	0.9	0.7	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.4	49.9	58.2	71.5	80.4	83.1	87.5	86.2	79.4	70.7	55.5	52.8		
		MCWB	39.3	39.5	43.5	51.8	55.5	63.5	67.3	70.6	61.2	55.6	43.7	40.3		
	2%	DB	43.1	45.1	51.9	66.3	74.4	77.5	82.0	80.7	74.1	62.7	49.2	46.6		
		MCWB	36.8	36.8	40.6	49.2	55.6	59.7	66.0	67.3	59.4	50.5	41.5	38.3		
	5%	DB	39.1	41.2	48.0	61.7	70.6	73.5	78.6	77.3	70.0	57.7	44.8	42.6		
		MCWB	34.1	34.8	38.7	46.5	53.8	57.6	64.5	64.7	57.8	47.7	38.3	36.4		
	10%	DB	35.1	37.8	43.9	56.8	66.2	70.0	75.3	74.0	65.7	53.8	40.5	38.7		
		MCWB	31.7	32.4	36.8	44.1	51.2	56.6	63.2	63.6	56.0	45.6	35.5	33.6		
	0.4%	WB	40.3	40.4	45.4	53.4	60.0	65.6	70.4	71.9	65.1	57.2	45.7	42.0		
		MCDB	46.3	48.5	56.4	67.8	74.4	77.8	80.4	80.6	73.3	69.3	54.2	50.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.4	37.3	41.7	50.3	57.6	62.8	68.0	68.3	61.8	52.0	41.9	38.9		
		MCDB	42.4	44.1	50.1	64.4	69.5	73.4	78.2	77.9	71.0	61.1	48.2	45.2		
	5%	WB	34.9	35.0	39.3	47.5	55.4	60.7	66.2	66.3	59.1	49.1	39.1	36.5		
		MCDB	38.8	40.5	46.6	59.7	67.1	70.0	76.6	74.9	67.4	56.2	44.4	42.1		
	10%	WB	31.6	32.7	37.1	44.9	53.2	58.8	64.3	64.3	56.7	46.6	36.1	33.7		
		MCDB	35.2	37.5	43.3	55.2	63.9	66.6	73.4	71.9	64.6	52.9	39.8	38.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.5	17.5	17.3	19.8	19.9	18.1	18.9	19.5	19.3	16.9	14.0	15.2		
		MCDBR	18.4	21.5	22.1	27.3	26.5	25.2	24.6	25.1	27.3	23.3	17.9	16.0		
		MCWBR	13.7	15.4	13.4	13.8	12.3	11.2	11.4	13.4	14.7	13.4	11.5	10.9		
		MCDBR	20.1	20.9	21.5	25.5	23.4	21.7	23.2	24.0	25.2	22.8	17.9	16.8		
	5% WB	MCWBR	16.0	15.3	13.7	14.0	11.3	11.2	12.3	14.0	14.5	14.1	12.3	12.3		
(40) Clear-Sky Solar Irradiance	taub		0.249	0.261	0.295	0.332	0.346	0.354	0.355	0.361	0.314	0.283	0.256	0.240		
	taud		2.346	2.385	2.392	2.379	2.382	2.375	2.426	2.362	2.494	2.560	2.464	2.319		
	Ebn at Noon		239	274	285	286	286	284	281	273	276	264	238	218		
	Edh at Noon		17	23	29	34	36	36	34	34	26	20	16	15		
(44) All-Sky Solar Radiation	RadAvg		304	628	1094	1470	1765	1881	1922	1590	1104	634	317	220		
	RadStd		23	48	69	134	128	127	117	108	113	68	30	19		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page